

**SECTION A**

General Unit Information:

Unit Name	Mathematics And Concepts For Computational Thinking - COPY
Unit Code	CIS093-1
Level	4
Credit Value	30
Location(s) of Delivery	On Campus and Off Campus
Period(s) of Delivery	13 weeks
Pre-requisites or Restrictions	N/A
Aims and Relevance	Computational thinking is a basis for solving computer related problems. This unit help you to develop the concepts of the computational thinking through converting theoretical knowledge of mathematics into practical skills of decomposition, pattern recognition, abstraction and algorithms which considered as the cornerstones of the thinking. The concepts of the computational thinking will be applied to implement suitable solutions for developing software applications using a high-level programming language (such as Python).
Syllabus Content	Data representation (including binary numbers) Concepts of mathematics that are relevant to computer science (including Boolean algebra, vectors and matrices) Algorithms Modelling using problem abstraction Program development using a high-level language Program testing using appropriate techniques

Learning Outcomes

On completion of this unit you should be able to

1. Demonstrate the following knowledge and understanding

Comprehend the mathematics that underpins computing problems and the techniques of decomposition, pattern recognition, abstraction and algorithms that are used in programming.

2. Demonstrate the following skills and abilities

Develop algorithms and software based on computational thinking using a high-level language following the stages of analysis, design, implementation and testing.

PSRB Outcomes and/or Apprenticeship Standard

N/A

Summary Learning Hours

	Scheduled	Guided	Independent	Autonomous	Placement	Total
Hours	72	108	90	30	0	300
Percentage	24%	36%	30%	10%	0%	100%

Approach to Learning

This unit is practice-based, combining lectures with practical exercises. For some of the practical exercises you will work in groups to encourage collaborative learning. The exercises will include the use of computational tools on a computer.

Assessment Summary

No	Assessment Method	Learning Outcome(s)	Weight	Submission Week	Length of Assessment
1	Summative in-class test or phase test	1 and 2	30	6	1 hour
2	Coursework - Artefact	1 and 2	70	13	2000 words

Learner Development

This Unit particularly focuses on the development of your abilities in the following areas:

Enquiry	You will develop the ability to select suitable methods and to apply them to computing tasks. You will evaluate the suitability and performance of the methods by carrying out computational experiments and analysing their results.
Contextual Understanding, including where employer based	You will understand the context in which the mathematics and concepts of computational thinking are developed and how they are applied to real-world problems. You will be encouraged to access and critically review mathematical texts to assist you in solving given problems.
Collaboration	You will be encouraged to learn collaboratively, to assist each other in comprehending and contextualising mathematical and statistical concepts and to critically assess and give feedback on the work of your peers.
Enterprise	You will have an opportunity to use the developed knowledge and skills for creatively approaching current computing challenges in an industry-wide context.

Assessment 1 Details	You will undertake a 1 hour in-class test which will cover core mathematical concepts relating to computing (such as boolean algebra and binary number systems).
Assessment 2 Details	You will submit an artefact (piece of computer software) as specified in the format outlined in the detailed assessment brief for the unit which will relate to the principles that underpin computing systems.
Assessment 3 Details	
Assessment 4 Details	

Threshold Expectations

In order to pass Assessment 1 you will need to :

Present solutions to a series of mathematical problems
Explain an example of a high-level abstraction concept or technique

In order to pass Assessment 2 you will need to:

Contribute to a software-based project in which you have a clearly defined role that involves writing a piece of software
Explain a design approach that solves a given computational problem
Conceptualise a computer program that compiles without any errors

In order to pass Assessment 3 you will need to:

In order to pass Assessment 4 you will need to:

Section B

Recommended Reading

Core Text - this unit is supported by the following core text:

See <https://rl.talis.com/3/bedfordshire/lists/B2028FC5-274C-1C24-2A88-955819606E6D.html?lang=en-GB>

Guided Reading

The following is expected reading for this unit. Details of what to read and when will be provide in the BREO site.

See <https://rl.talis.com/3/bedfordshire/lists/B2028FC5-274C-1C24-2A88-955819606E6D.html?lang=en-GB>

Independent Study

To receive high grades you will need to demonstrate your wider reading. The following resources provide useful background reading for the material in this unit. This is not an exhaustive list and students should read widely from the variety of journals available in the Learning Resources Centre.

See <https://rl.talis.com/3/bedfordshire/lists/B2028FC5-274C-1C24-2A88-955819606E6D.html?lang=en-GB>

Equality Impact Assessment

Question	Y/N/NA	Additional anticipatory adjustments/actions if necessary
Learning materials will be made available in advance of sessions for students to adapt as appropriate?	Y	
The approach to teaching and learning is sufficiently flexible to enable all students to succeed?	Y	
The approach to group work takes account of the needs of students with disabilities and from diverse backgrounds?	NA	
The approach to practical work takes account of the needs of students with disabilities?	N	Students diagnosed with Developmental Dyscalculia may find this unit exceptionally difficult. They will be referred to Student Support to identify the best action forward.
Students with a protected characteristic* have an equal opportunity to achieve the learning outcomes?	Y	
The assessment tasks provide all students with an equal opportunity to succeed?	Y	

Any other aspects of the unit that might pose potential challenges from an equality or diversity perspective have been considered?	NA	
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* Age, Gender reassignment, Marriage and civil partnership, Pregnancy and maternity, Race, Religion and belief, Sex, Sexual orientation

Section C

Administrative Information – Faculty completion	
Faculty	Creative Arts Technologies and Science
Portfolio	CU-COM
School/Department	School of Computer Science and Technology
HECOS Code(s)	100374
Unit Co-ordinator	Susan Brandreth
Apprenticeship ?	No

Courses that this unit is delivered on
BSc (Hons) Computer Science
BSc (Hons) Software Engineering
BSc (Hons) Cybersecurity
BSc (Hons) Computer Networking
BSc (Hons) Business Information Systems
BSc (Hons) Information and Data Systems
BSc (Hons) AI & Robotics
BSc (Hons) Cybersecurity - Hong Bang International University (HBIU)
BSc (Hons) Cybersecurity - Middle East University (MEU)